

06 Vector Space

6.1 Vector Space

The natural extension of the Cartesian coordinate system — 8 axioms, linear combinations, linear dependence, span, rank, basis, and dimension



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Build

1

A vector space is built from vector addition and scalar multiplication, and must satisfy 8 basic axioms

3

Linear dependence and independence: if some vector can be written as a combination of the others, the set is dependent; otherwise, independent

5

Maximal linearly independent subset and rank: the largest linearly independent subset of a vector group; its size is the rank

7

Dimension $\dim()$: the number of vectors in a basis

2

Linear combination: a weighted sum of a set of vectors, producing a new vector

4

Span: the set of all linear combinations of a group of vectors

6

Basis: a linearly independent set of vectors that spans the space, uniquely representing every vector in it

8

Equivalences and affine space: a square matrix is full rank $\Leftrightarrow$ invertible $\Leftrightarrow$ has independent columns (rows) $\Leftrightarrow$ has nonzero determinant; translating a vector space gives an affine space

From Coordinate Systems to Vector Spaces

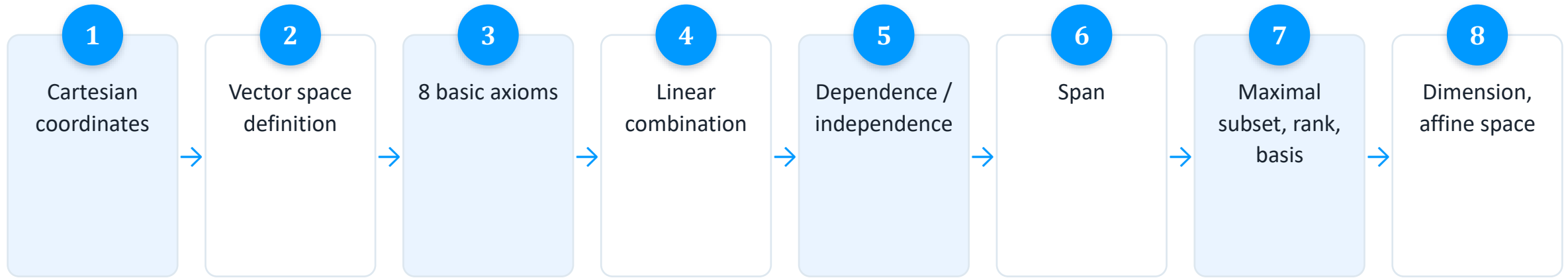
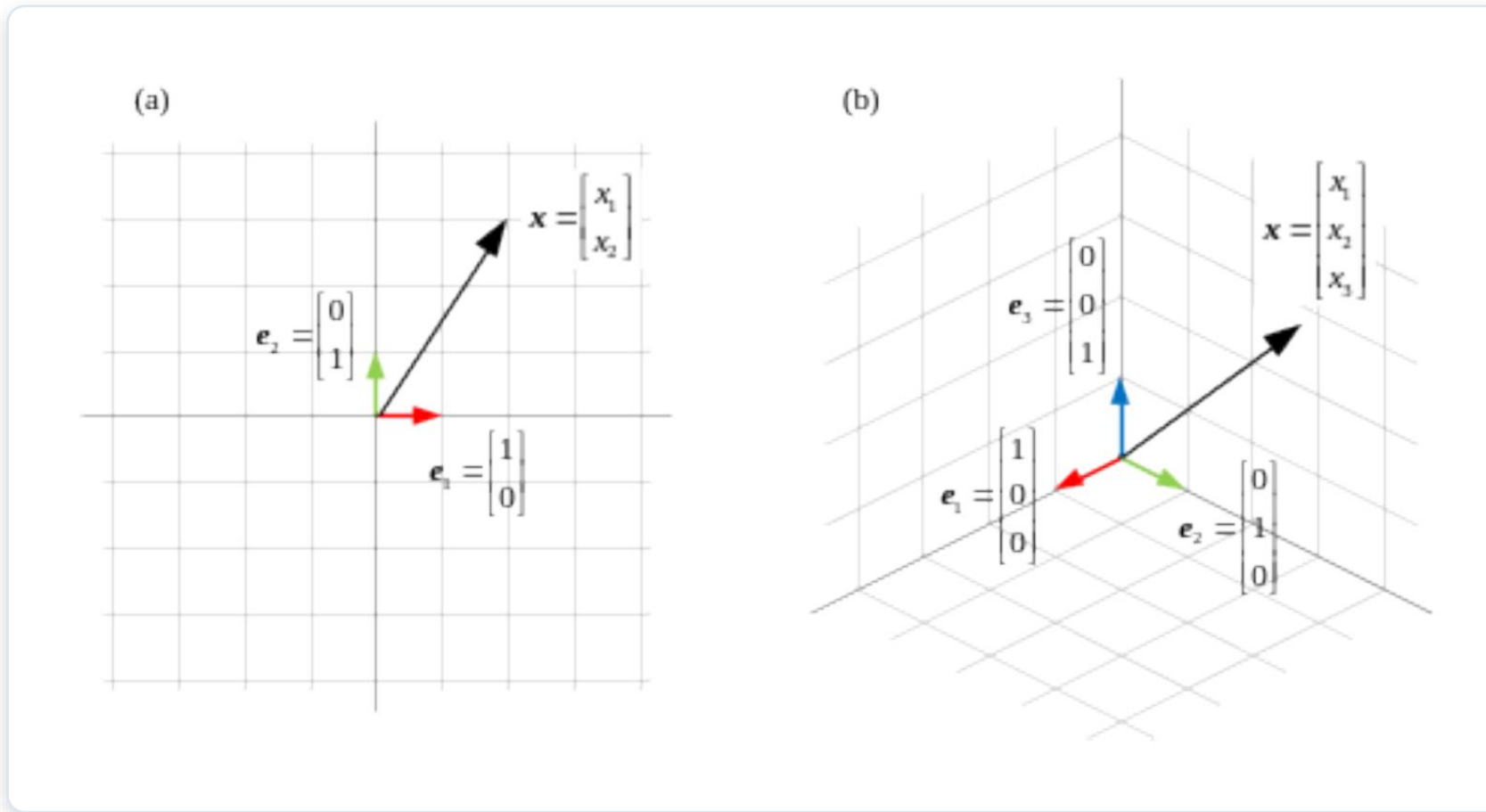


Figure 1 · 2D and 3D Cartesian Coordinate Systems

- $\mathbb{R}^2$ and $\mathbb{R}^3$ are the most basic Euclidean vector spaces; $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ act like a “skeleton” holding up the whole space



The Precise Definition of a Vector Space

$$\mathbf{a}, \mathbf{b} \in V \Rightarrow \mathbf{a} + \mathbf{b} \in V, k\mathbf{a} \in V$$

- Given a field F , a vector space V over F is a nonempty set closed under addition and scalar multiplication
- V , together with these two operations, is called a vector space if it satisfies 8 axioms
- No need to memorize the axioms — this section uses the RGB color space as an intuitive analogy

Figure 2 · $a + b = b + a$

- In RGB color space, red+blue and blue+red both give magenta — commutativity holds visually

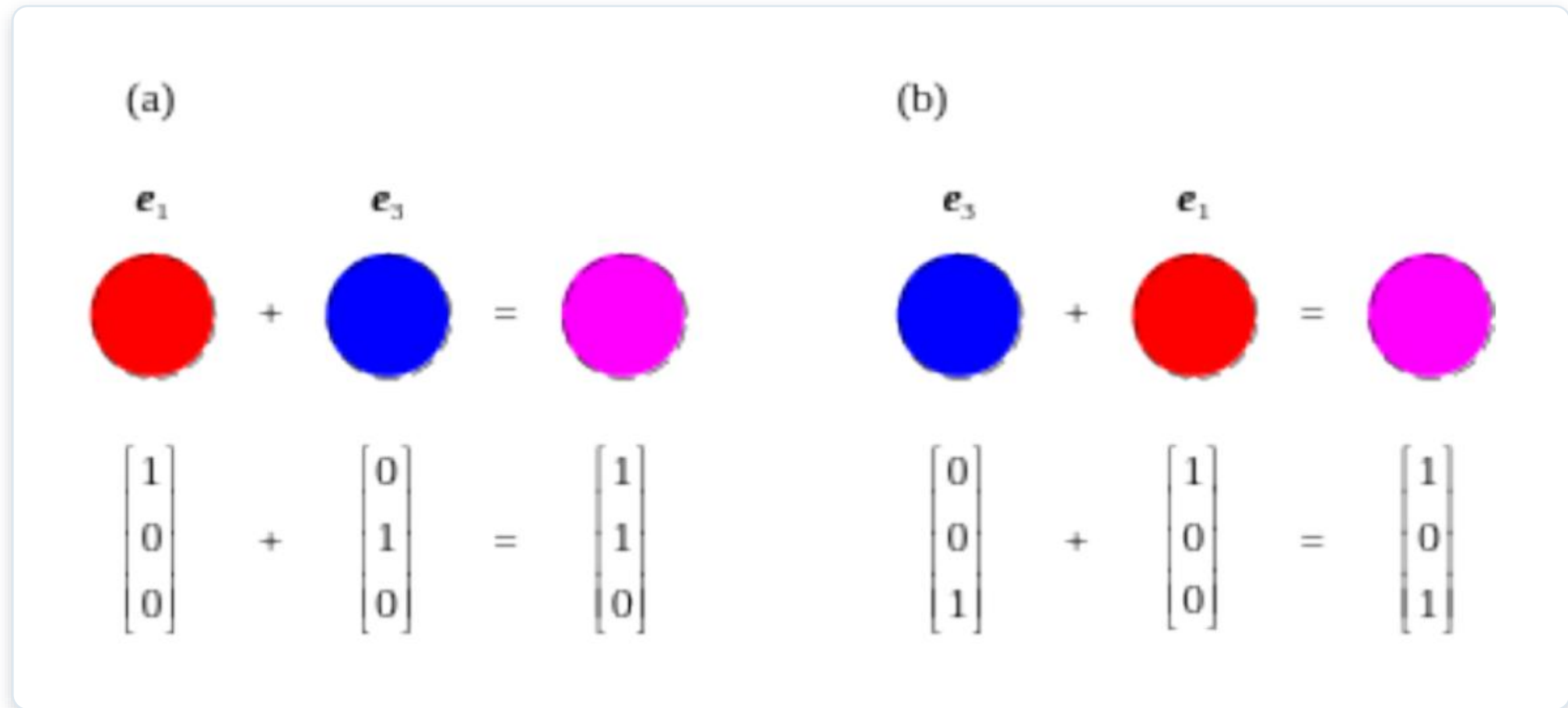


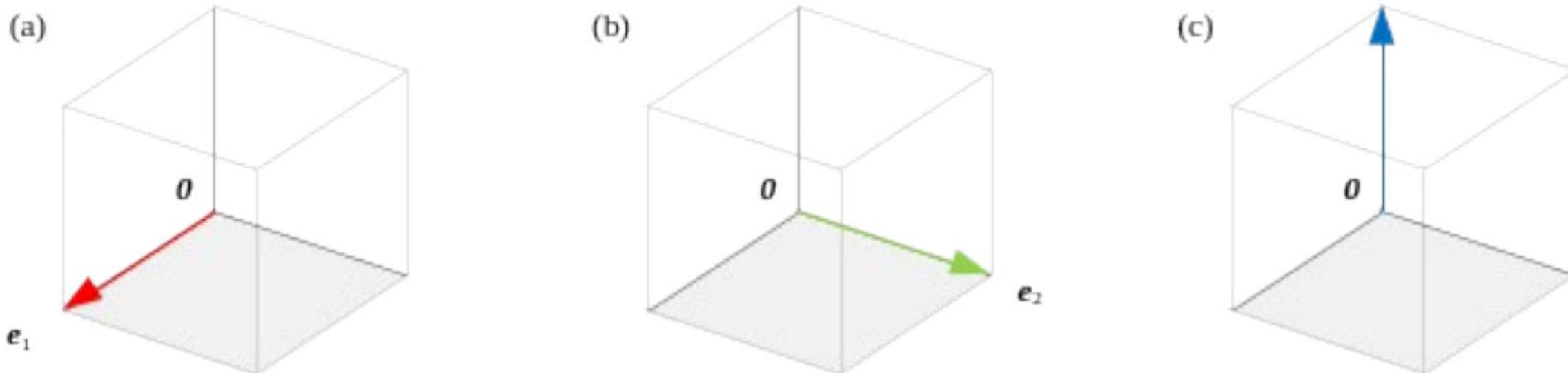
Figure 3 · $(a+b)+c = a+(b+c)$

- Red+green first then +blue, or green+blue first then +red — either path lands on white



Figure 4 · $a + 0 = a$

- The zero vector 0 is like black (no light): adding black to any color leaves it unchanged



AXIOM 4 · EXISTENCE OF ADDITIVE INVERSE

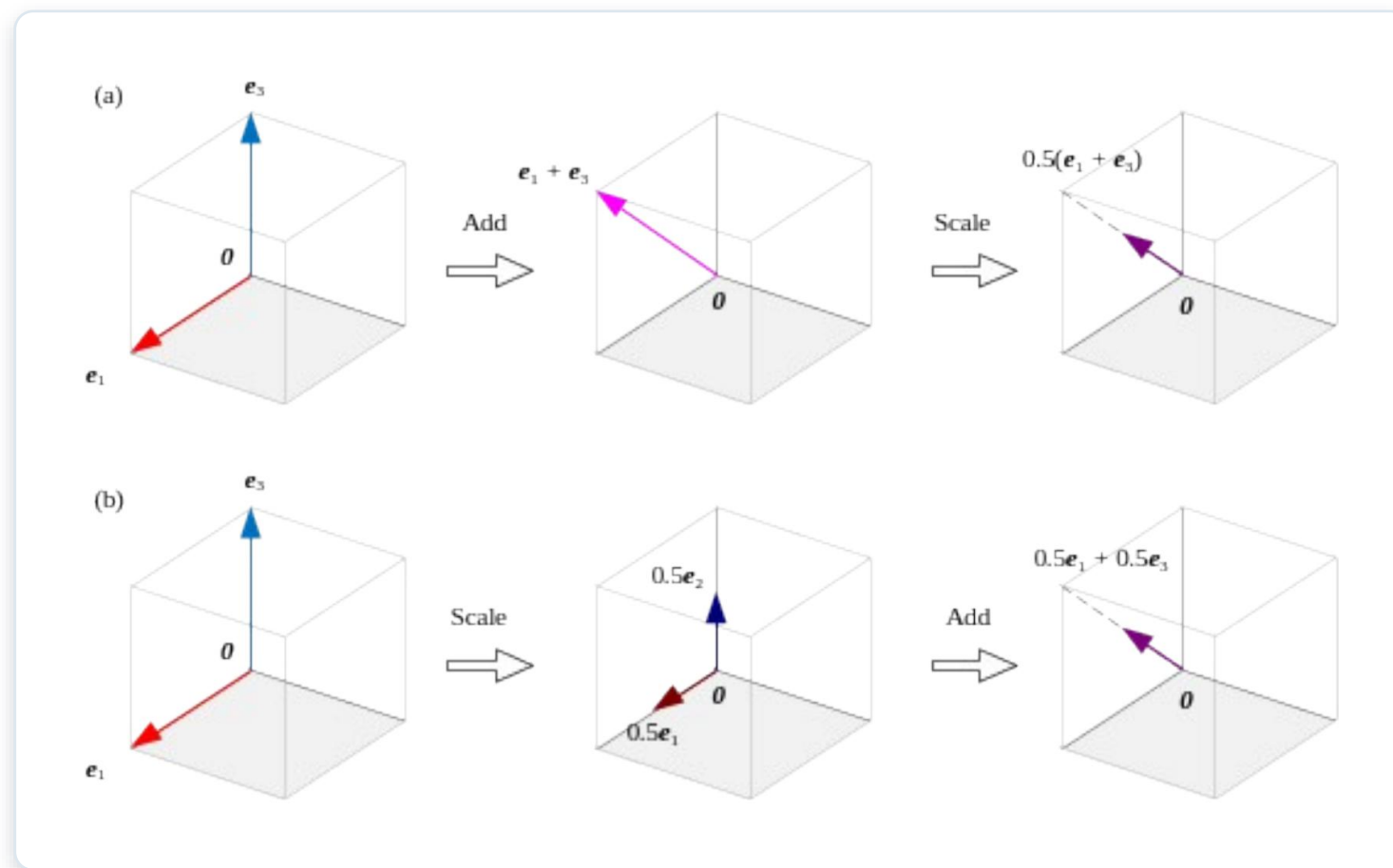
$$\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$$

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- For every element $\mathbf{a}$ in V , there exists another element $-\mathbf{a}$ such that $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$
- RGB has no true negative color (components are clamped to $[0,1]$), but $-\mathbf{a}$ can be read as “removing” a color
- Removing red light from red light itself leaves black — the zero vector $\mathbf{0}$

Figure 5 · $k(a+b) = ka + kb$

- Mix e_1, e_3 first then scale by 0.5, or scale each by 0.5 first then add — same result



$$(k + t)a = ka + ta$$

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- For any scalars k, t and any element a in V , $(k+t)a = ka + ta$
- Boosting orange to 0.5 intensity: combine $0.3+0.2=0.5$ then scale once, or scale twice by 0.3 and 0.2 then add — same result

$$(kt)a = k(ta)$$

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- For any scalars k, t and any element a in V , $(kt)a = k(ta)$
- Shrinking cyan by 0.2 then by 0.5 gives the same result as shrinking it directly by 0.1 (0.2×0.5)

AXIOM 8 · SCALAR MULTIPLICATION IDENTITY

$$\mathbf{1} \cdot \mathbf{a} = \mathbf{a}$$

$$1 \cdot \mathbf{a} = \mathbf{a}$$

- For any element $\mathbf{a}$ in V , $1 \cdot \mathbf{a} = \mathbf{a}$
- Multiplying any color vector by 1 leaves the color unchanged

The 8 Axioms of Vector Addition and Scalar Multiplication

Axiom 1 Commutativity: $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$

Axiom 2 Associativity: $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$

Axiom 3 Additive identity: $\mathbf{a} + \mathbf{0} = \mathbf{a}$

Axiom 4 Additive inverse: $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$

Axiom 5 Distributivity: $k(\mathbf{a} + \mathbf{b}) = k\mathbf{a} + k\mathbf{b}$

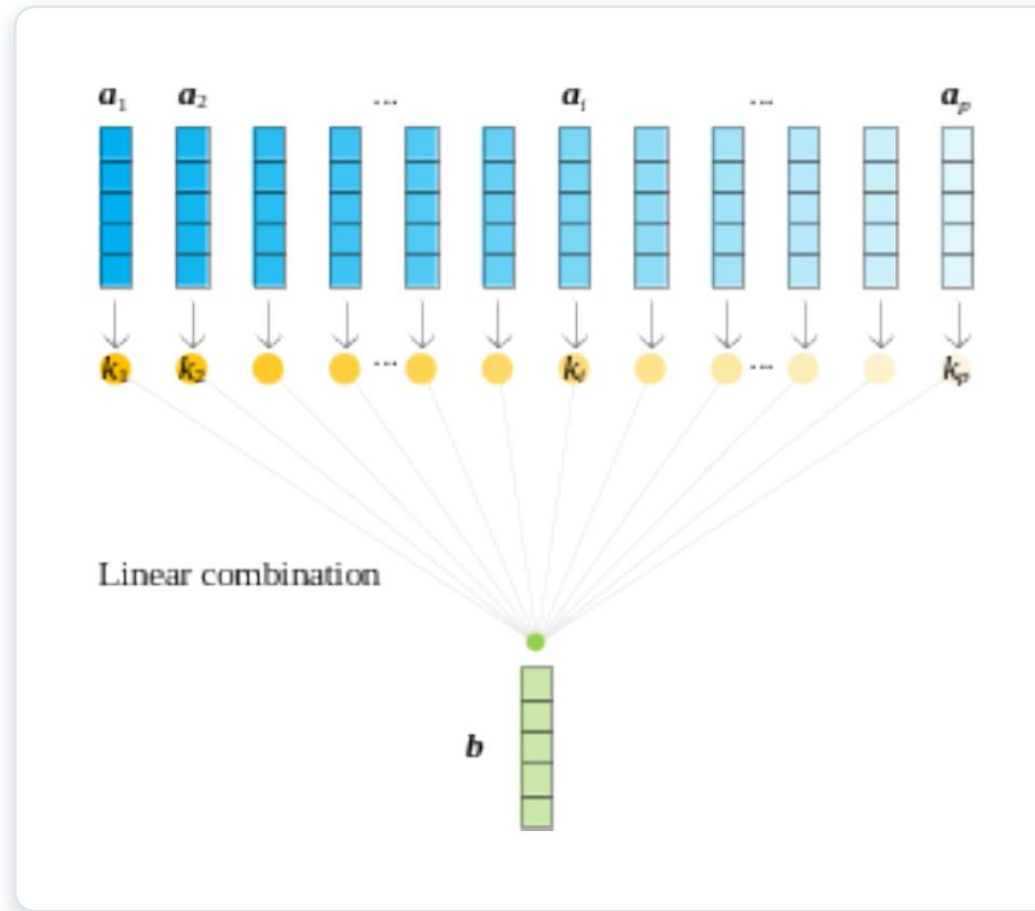
Axiom 6 Distributivity: $(k + t)\mathbf{a} = k\mathbf{a} + t\mathbf{a}$

Axiom 7 Scalar associativity: $(kt)\mathbf{a} = k(t\mathbf{a})$

Axiom 8 Scalar identity: $1 \cdot \mathbf{a} = \mathbf{a}$

Figure 6 · A Linear Combination of Column Vectors

- Given $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$ and weights $k_1, k_2, \dots, k_p$, $\mathbf{b} = k_1\mathbf{a}_1 + k_2\mathbf{a}_2 + \dots + k_p\mathbf{a}_p$



Defining Linear Dependence and Independence

$$k_1 \mathbf{a}_1 + k_2 \mathbf{a}_2 + \cdots + k_p \mathbf{a}_p = \mathbf{0}$$

- Given $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p]$, if there exist $k_1, \dots, k_p$, not all zero, satisfying the equation above, $\mathbf{A}$ is linearly dependent; otherwise independent
- Independence means no vector in the set can be written as a combination of the others — they each contribute their own direction

Figure 7 · Linearly Dependent Groups in RGB Space

- In (a), $[e_1, e_2, e_1 + e_2]$ is dependent since $e_1 + e_2$ is a combination of e_1, e_2 ; (b)(c) follow the same logic

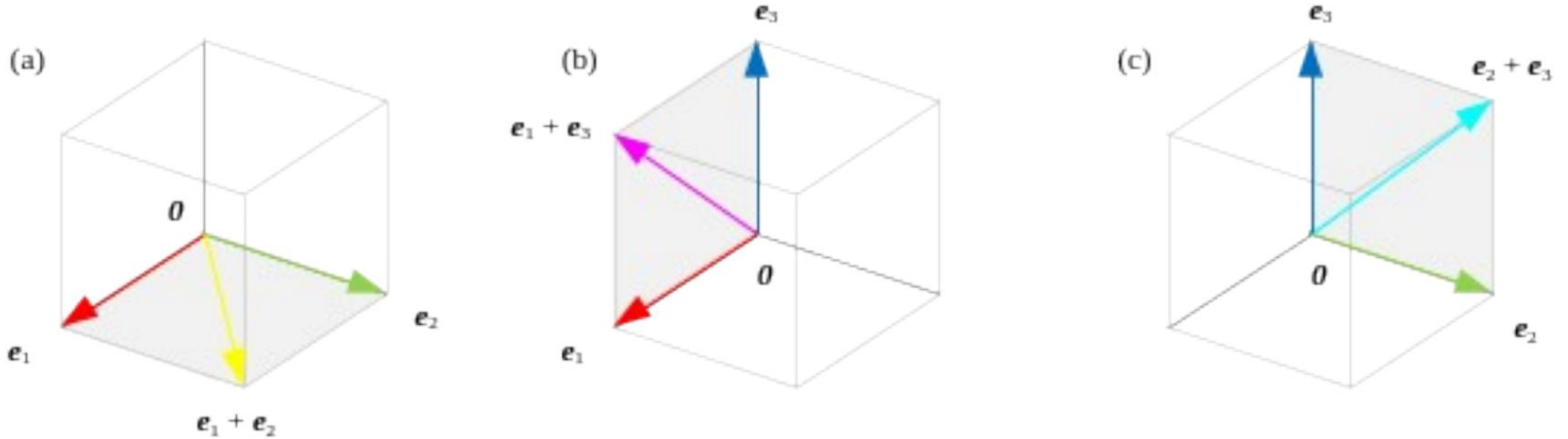


Figure 8 · Dependent Groups with More Than Three Vectors

- Whenever the vector count exceeds the space's dimension, the group must be dependent — any 4 vectors in 3D are dependent

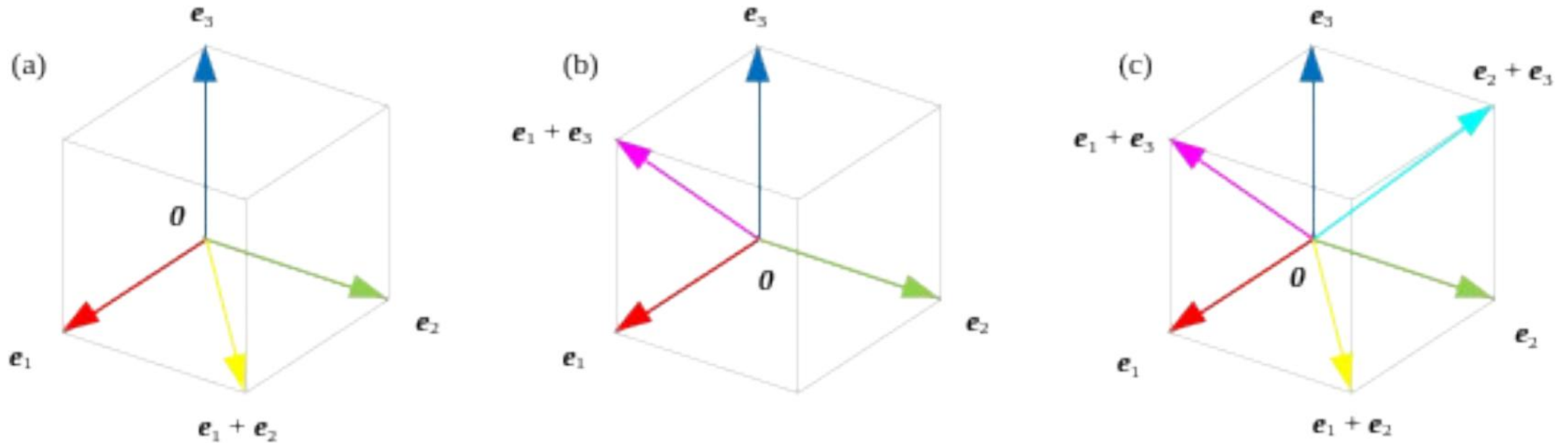


Figure 9 · Linearly Independent Groups in RGB Space

- $\mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{e}_1$ each contribute a distinct direction and can't replace one another — independent even when not orthogonal

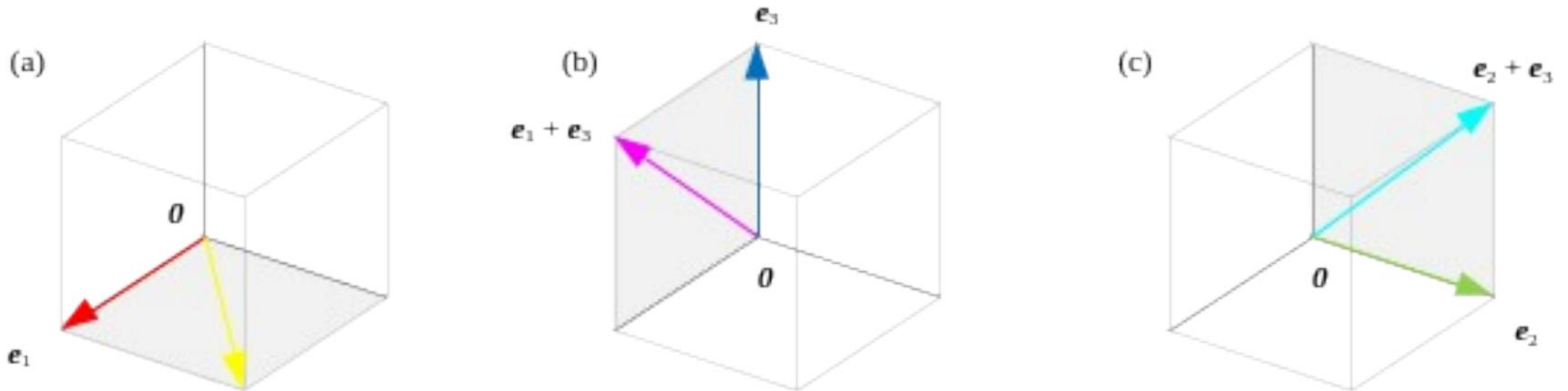
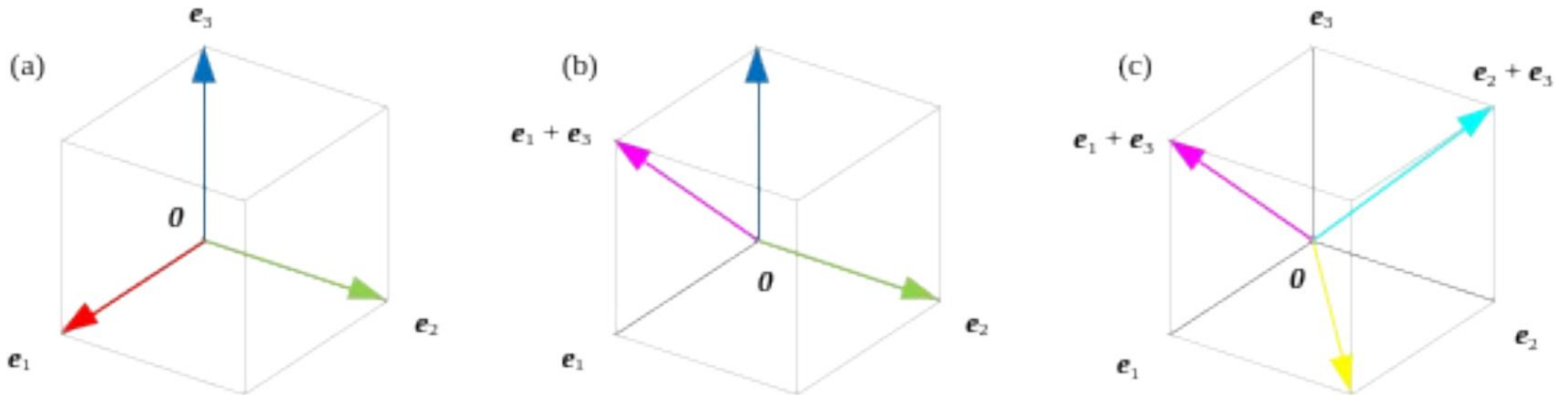


Figure 10 · Linearly Independent Groups in 3D Space

- 3D space needs at least three “skeleton” vectors, and they need not be pairwise orthogonal — a preview of basis



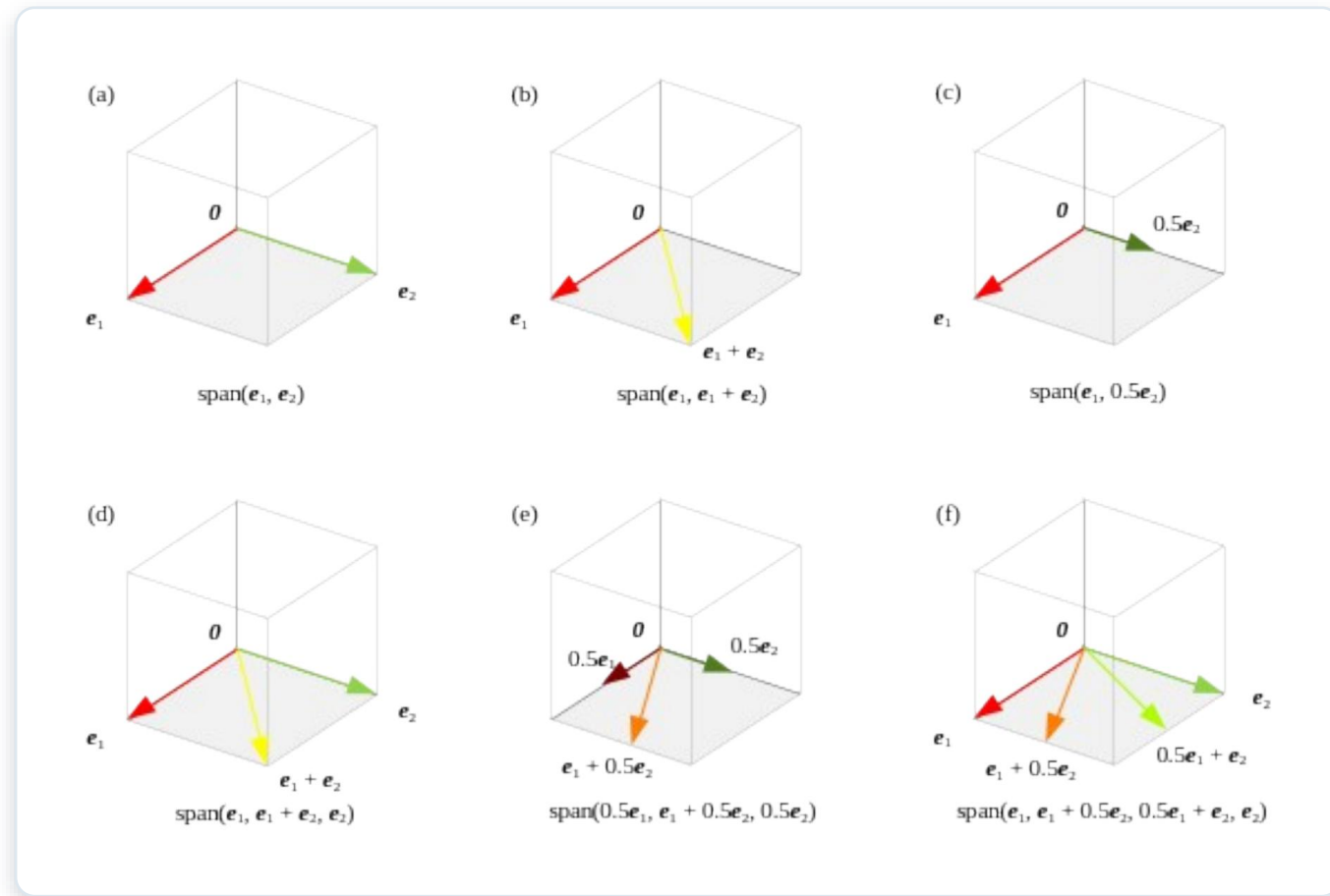
Defining the Span, `span()`

$$V = \text{span}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p)$$

- The set of all linear combinations of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$ is called their span, written $\text{span}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p)$
- For a span, the vectors may be dependent or independent — either way they can span the same space

Figure 11 · Six Groups Spanning the Same “Red-Green” Plane

- (a)(b)(c) are themselves independent, (d)(e)(f) are themselves dependent, yet all six span the same plane



Maximal Linearly Independent Subset and Rank

$$\text{rank}(\mathbf{A}) = \text{column rank} = \text{row rank}$$

- The column rank of $\mathbf{A}$ is the largest number of linearly independent columns; row rank is defined similarly, and the two are always equal — together called $\text{rank}(\mathbf{A})$
- If $\mathbf{A}$'s columns are dependent, redundant vectors can be removed one by one until independent, giving a maximal linearly independent subset — its size is the rank
- Note: a maximal linearly independent subset is not unique

Figure 12 · Removing Redundant Vectors from Figure 7(a)

- Removing any one vector from $[e_1, e_2, e_1 + e_2]$ leaves two vectors that form a maximal independent subset

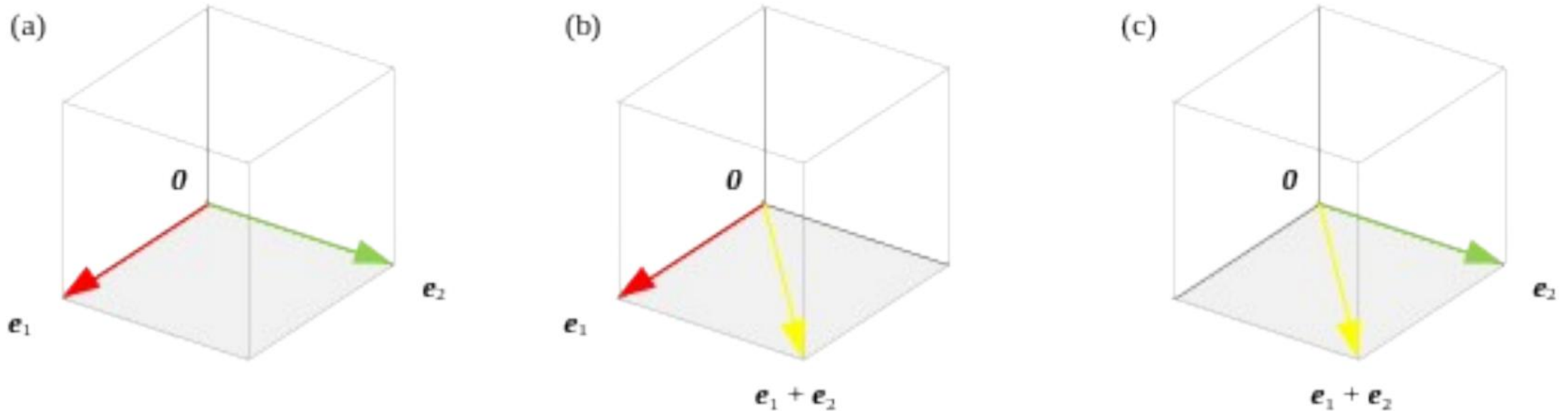
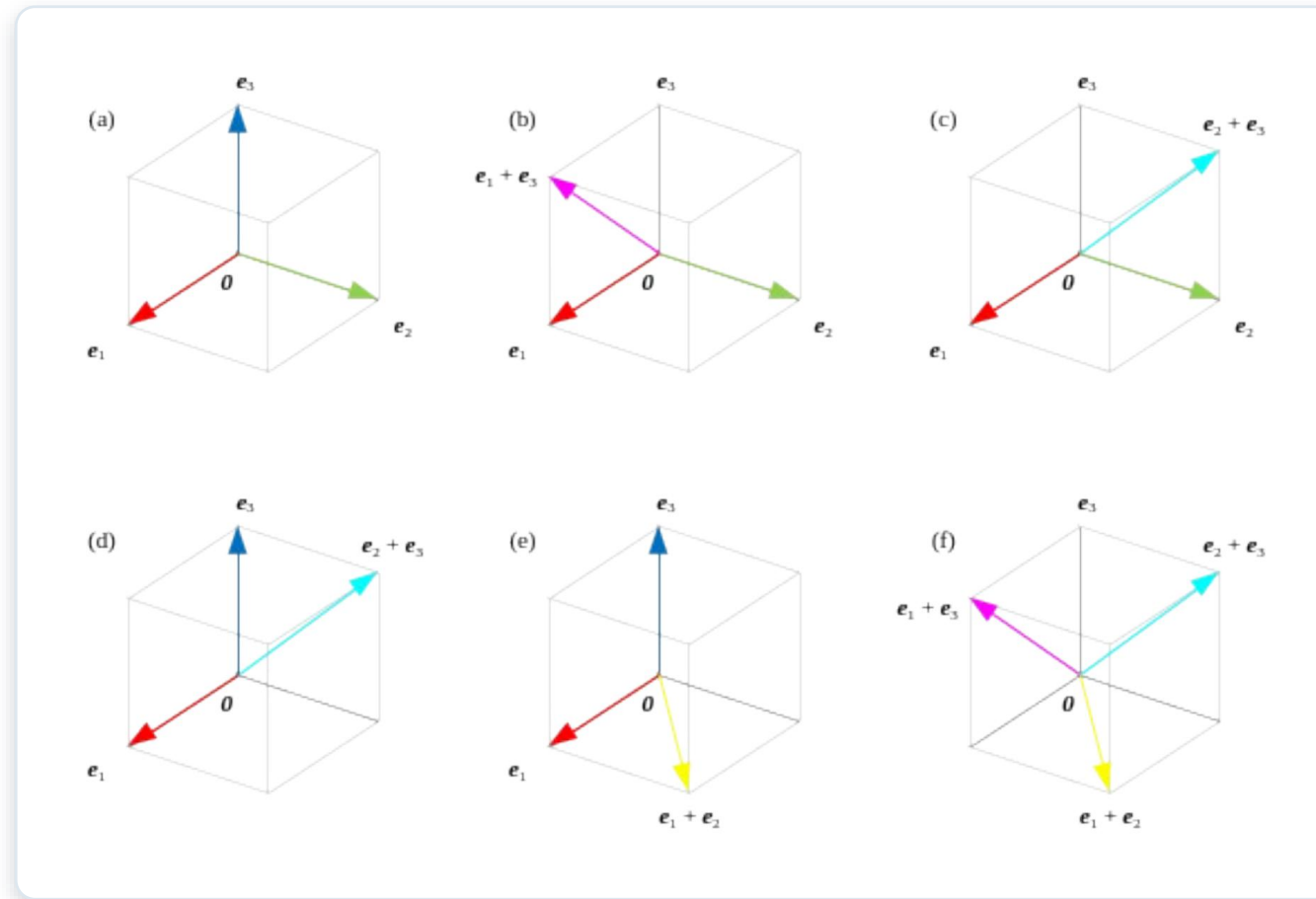


Figure 13 · Removing Redundant Vectors from Figure 8(c)

- Each vector group in Figure 13 still spans the entire RGB color space — each is a maximal independent subset



Code 1 · Computing the Rank of a Matrix

Code 1. Computing Rank |
Bk1_Ch06_01_01.ipynb

- ❶ Define three column vectors $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ with `np.array()`
- ❷ Stack them into matrix $\mathbf{A}$ with `np.column_stack()`, then call `np.linalg.matrix_rank()`
- Matrix $\mathbf{B} = [\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ has rank 2, since $\mathbf{e}_1 + \mathbf{e}_2$ is redundant
- Matrix $\mathbf{D} = [\mathbf{e}_1, -\mathbf{e}_1, \mathbf{e}_1/2]$ has rank 1 — all three columns lie on one line

```
## 初始化
import numpy as np

## 定义列向量
a e_1 = np.array([[1],[0],[0]])
  e_2 = np.array([[0],[1],[0]])
  e_3 = np.array([[0],[0],[1]])

## 计算秩
b A = np.column_stack([e_1, e_2, e_3])
  np.linalg.matrix_rank(A)

B = np.column_stack([e_1, e_2, e_1 + e_2])
  np.linalg.matrix_rank(B)

C = np.column_stack([e_1, e_2])
  np.linalg.matrix_rank(C)

D = np.column_stack([e_1, -e_1, e_1/2])
  np.linalg.matrix_rank(D)
```



Basis: The “Skeleton” That Holds Up a Space

$$V = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_p), \text{ with } \mathbf{a}_1, \dots, \mathbf{a}_p \text{ linearly independent}$$

- A basis (vector basis) is a linearly independent set of vectors that “holds up” the whole space
- Every vector in V can be written uniquely as a combination of the basis vectors
- A basis is like a map's longitude and latitude — it pins down location

$[e_1, e_2]$ Is Exactly the 2×2 Identity Matrix

$$[e_1, e_2] = I_{2 \times 2}, [e_1, e_2, e_3] = I_{3 \times 3}$$

- Distinguish $\{e_1, e_2\}$ (an unordered set) from $[e_1, e_2]$ (an ordered basis) — this book always treats bases as ordered
- $[e_1, e_2]$ and $[e_1, e_2, e_3]$ are called the standard basis; infinitely many bases span the same space

Dimension $\dim()$: the “Size” of a Space

$\dim(V)$ = the number of vectors in a basis

- The dimension of a vector space is the number of vectors in its basis, written $\dim()$
- $\mathbb{R}^2 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$, $\dim(\mathbb{R}^2) = 2$; $\mathbb{R}^3 = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$, $\dim(\mathbb{R}^3) = 3$

Figure 14 · Six Vector Spaces of Dimension 1

- Every one of these spaces is a line through the origin — $\mathbf{0}$ lies in each of them

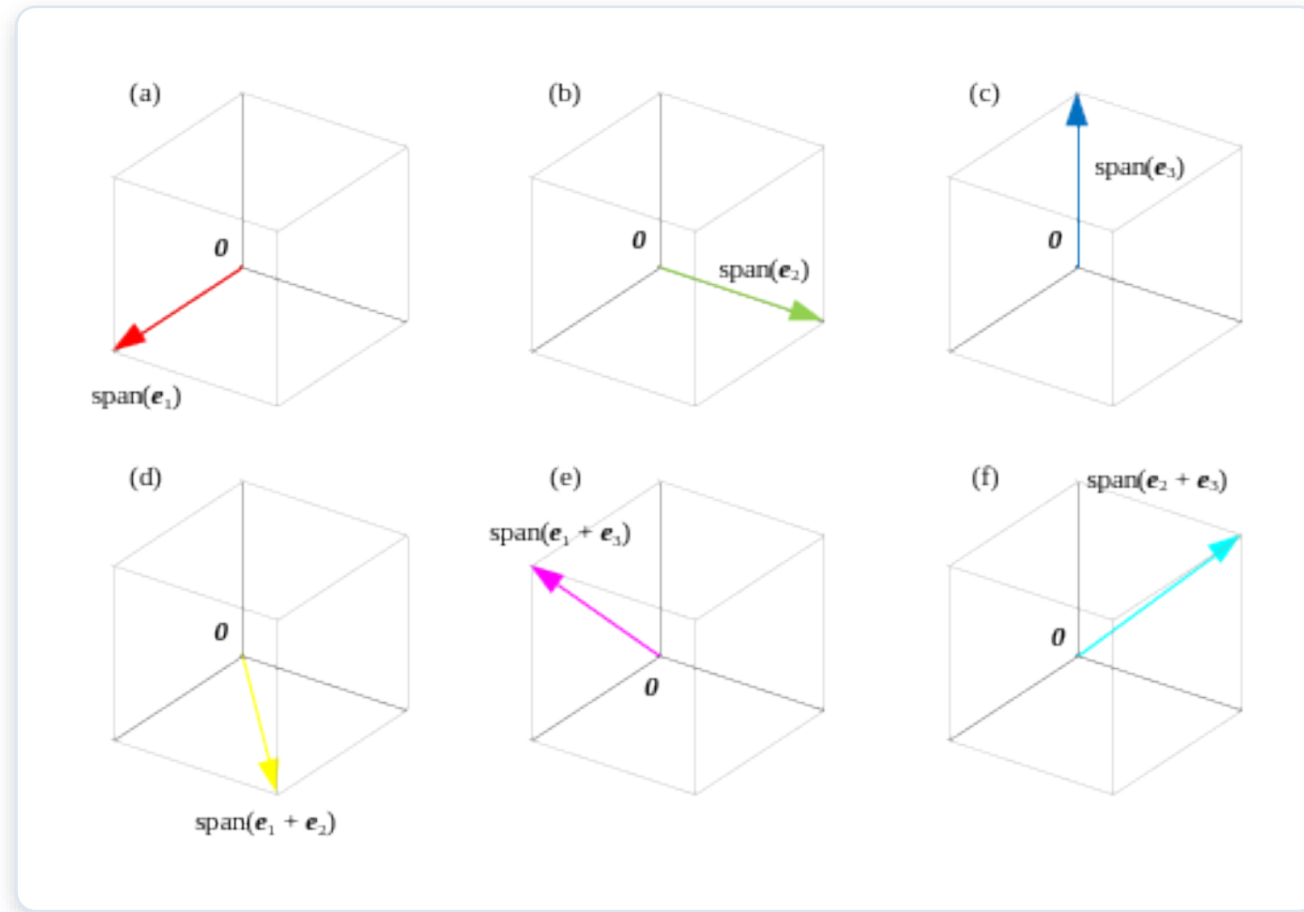


Figure 15 · 2D Spaces Spanned by Independent Vectors

- Each pair of vectors is a basis for that space; note $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$, since their vectors have different lengths

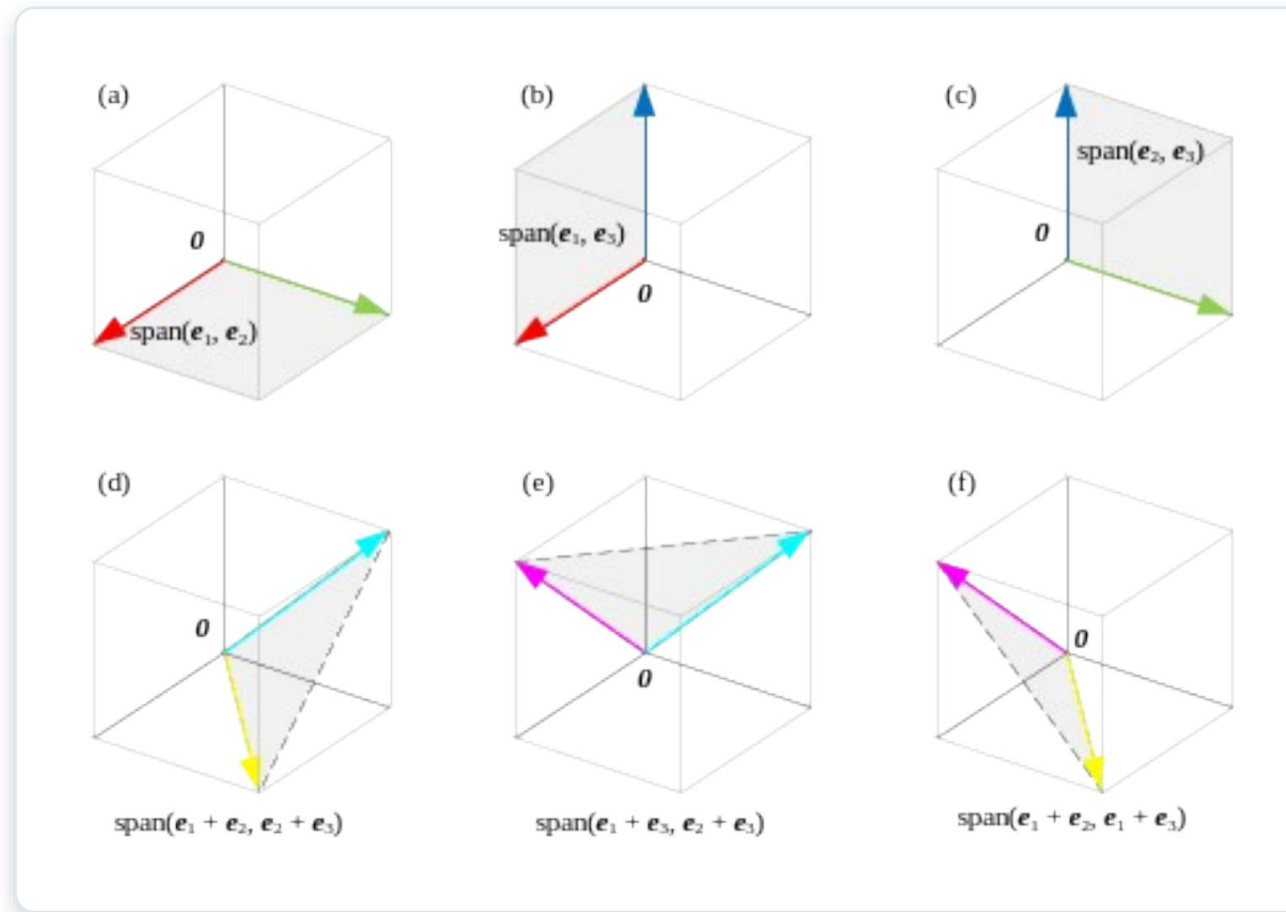


Figure 16 · 2D Spaces Spanned by Dependent Vectors

- $\text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2)$ has dimension 2, but $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ is dependent, so it can't be called a basis

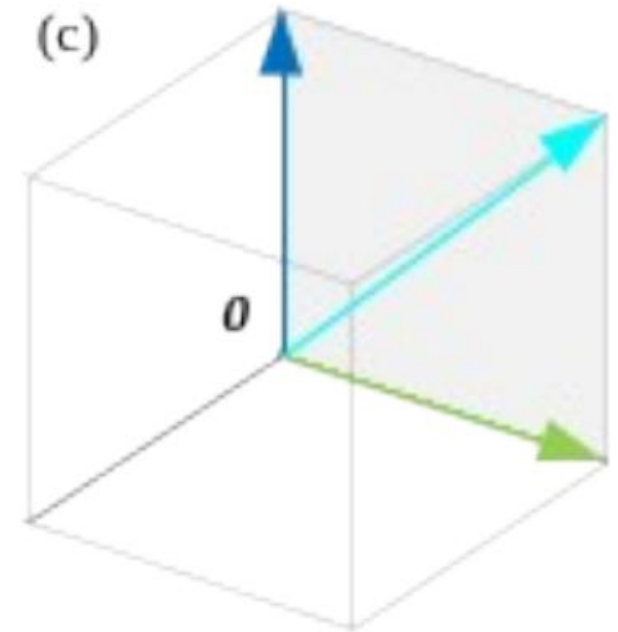
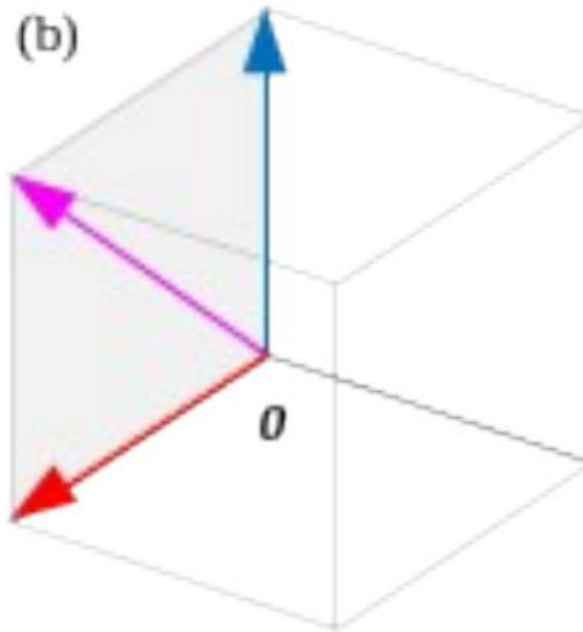
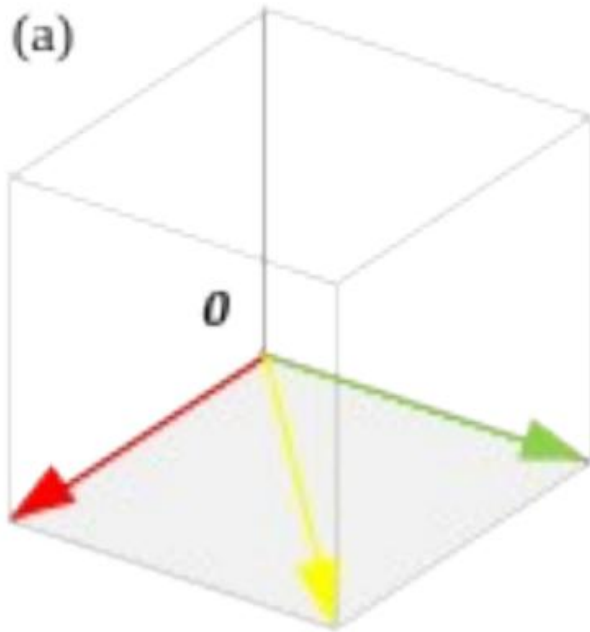
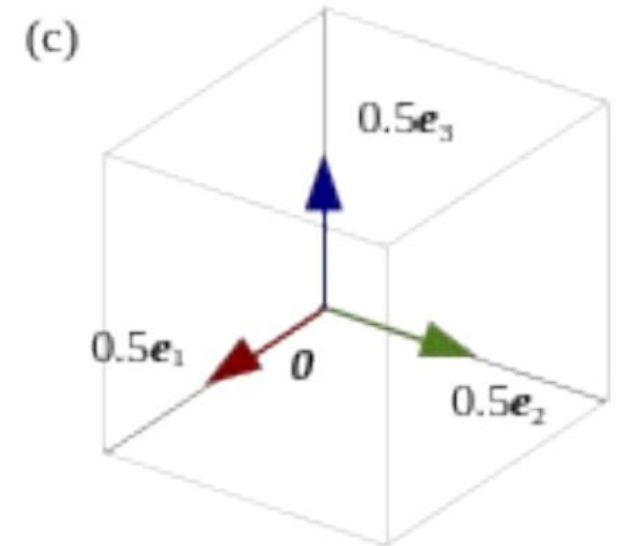
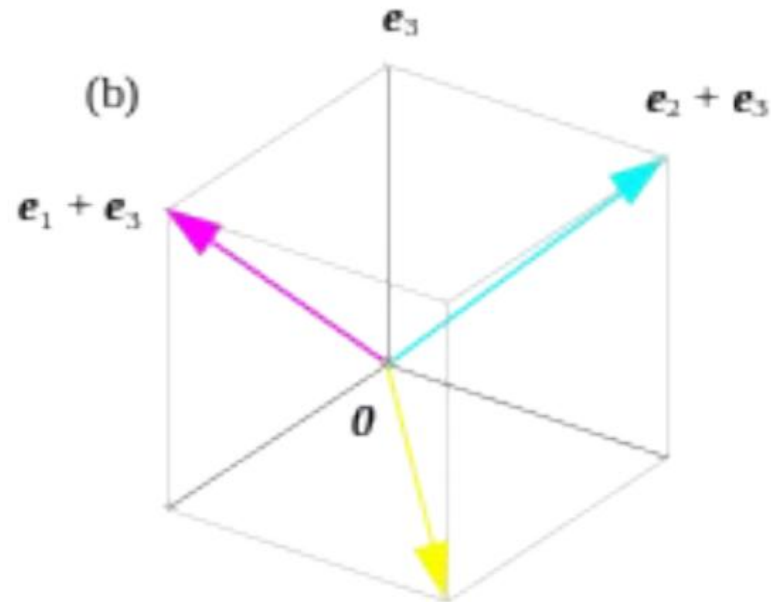
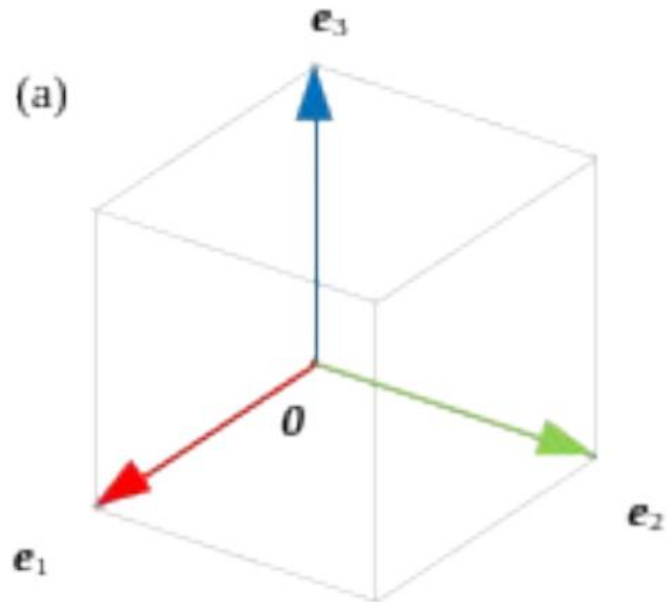


Figure 17 · 3D Spaces Spanned by Independent Vectors

- Every group shown is a basis for this space; basis vectors may or may not be orthogonal



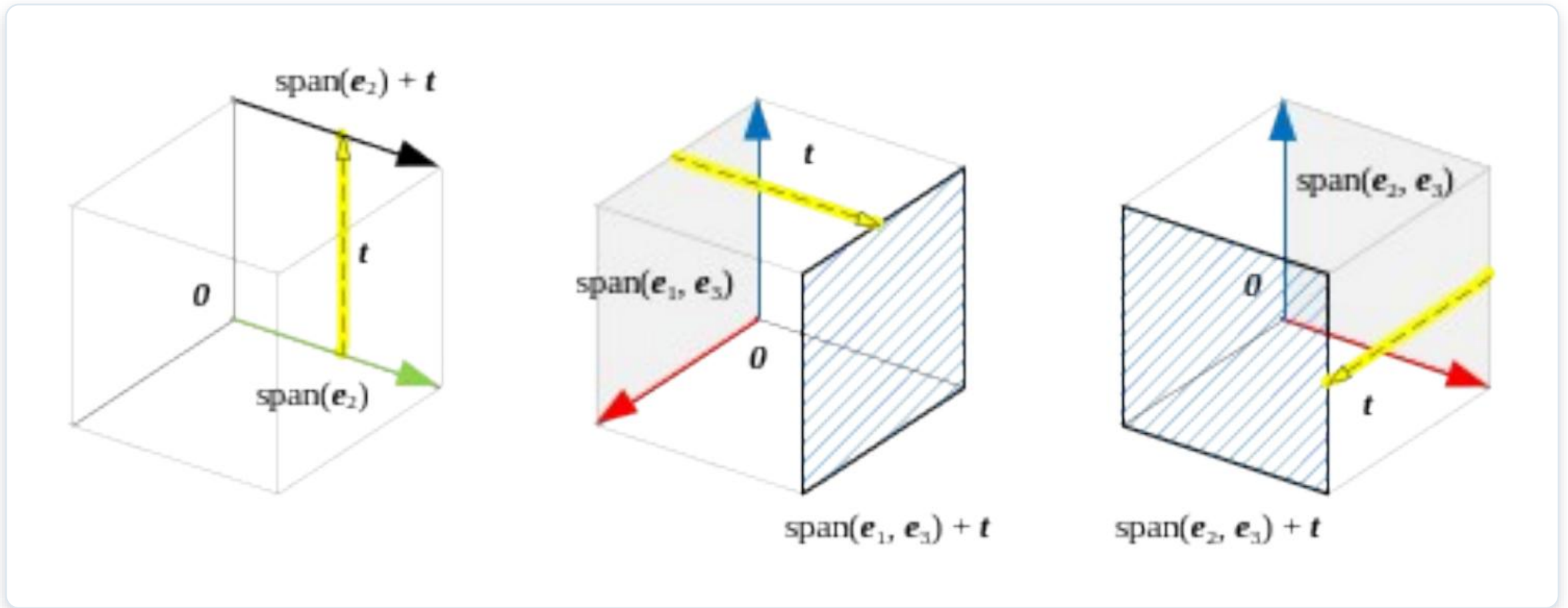
Translating a Vector Space Gives an Affine Space

$$\text{affine space} = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_p) + \mathbf{t}$$

- “Passing through the origin” matters a great deal — every space in Figures 14–17 contains $\mathbf{0}$
- A vector space translated away from the origin is called an affine space; it no longer contains the origin

Figure 18 · Translating a Vector Space into an Affine Space

- All three affine spaces are the vector space shifted along t — none of them passes through the origin



The Full Framework, from Coordinates to Vector Spaces

full rank $\Leftrightarrow$ invertible $\Leftrightarrow$ independent columns (rows) $\Leftrightarrow$ nonzero determinant

- A vector space is defined by addition, scalar multiplication, and 8 axioms; linear combination, dependence, span, rank, basis, and dimension all interlock
- A basis must be independent and must span the space; a space has infinitely many bases, but a single dimension
- Translating a vector space gives an affine space, which no longer contains the origin

Key Terms in This Section

Vector Space *A nonempty set closed under addition and scalar multiplication, satisfying 8 axioms*

Linear Dependence *There exist k 's, not all zero, with $k_1\mathbf{a}_1 + \dots + k_p\mathbf{a}_p = \mathbf{0}$*

Span *The set of all linear combinations of $\mathbf{a}_1, \dots, \mathbf{a}_p$, written $\text{span}(\mathbf{a}_1, \dots, \mathbf{a}_p)$*

Rank *The size of a maximal independent subset, $\text{rank}(\mathbf{A}) = \text{column rank} = \text{row rank}$*

Dimension *The number of vectors in a basis, $\dim(V)$*

Linear Combination $\mathbf{b} = k_1\mathbf{a}_1 + k_2\mathbf{a}_2 + \dots + k_p\mathbf{a}_p$

Linear Independence *No vector in the group can be written as a combination of the others*

Maximal Independent Subset *The largest linearly independent subset of a vector group*

Basis *A linearly independent set of vectors that spans a vector space*

Affine Space *A vector space after translation; no longer contains the origin*

GO FURTHER

Companion Code and Further Reading

Companion Code · Bk1_Ch06_01_01.ipynb

Uses `np.column_stack()` and `np.linalg.matrix_rank()` to compute rank and check dependence/independence

Further Reading · Subspaces

A subspace is a nonempty subset of a vector space closed under addition and scalar multiplication, containing the zero vector

Further Reading · Orthogonal and Non-Orthogonal Bases

Basis vectors may or may not be orthogonal; later chapters cover constructing orthogonal bases

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Q1 – Q8

Q1 Which of these are not a basis? $[e_1, e_2]$ / $[e_1, e_2, e_1 + e_2]$ / $[e_1, e_2, e_3]$ / $[e_2, e_3]$ / $[e_1, e_2, e_3, e_1 + e_2]$ / $[0.5e_1, e_2, 0.5e_3]$

Q3 How do dependent/independent columns of a square matrix relate to its invertibility?

Q5 How do dependent/independent columns of a square matrix relate to whether its determinant is zero?

Q7 Write `is_basis(vectors)`: given a set of column vectors, return True if they're independent and their count equals the space's dimension.

Q2 For each non-basis group in Q1, which vectors should be removed to form a basis?

Q4 How does a square matrix being full column rank / not full rank relate to its invertibility?

Q6 Use `np.column_stack()` and `np.linalg.matrix_rank()` to check the rank of every group in Q1, and decide which are bases.

Q8 Build the 3×3 matrix $[e_1, e_2, e_1 + e_2]$ with numpy and verify its rank is below 3, showing it's not a basis for $\mathbb{R}^3$.

Q9 – Q16

Q9 Use `numpy.linalg.det()` on the matrices $[e_1, e_2, e_3]$ and $[e_1, e_2, e_1 + e_2]$, and compare whether each determinant is zero.

Q11 Plot the three vectors $[e_1, e_2, e_1 + e_2]$ from Figure 7(a) with matplotlib, using a distinct color for each.

Q13 Verify with numpy that $(k+t)a = ka + ta$ (Axiom 6) holds for arbitrary scalars k, t and vector a .

Q15 Sweep RGB channel combinations in code to test whether RGB space satisfies Axiom 5, and explain why RGB isn't a true vector space.

Q10 Using matplotlib's 3D tools, draw two independent vectors and the plane they span, following Figure 9.

Q12 Write `span_check(v, basis)` using `numpy.linalg.lstsq()` to test whether vector v lies in the span of basis .

Q14 Verify with numpy that $(kt)a = k(ta)$ (Axiom 7) holds for arbitrary scalars k, t and vector a .

Q16 Construct 4 vectors in 3D with numpy and verify they must be dependent (more vectors than the space's dimension).

Q17 – Q25

Q17 Write `maximal_independent_subset(vectors)` to pick out a maximal linearly independent subset from a given group.

Q19 Build the affine space $\text{span}(\mathbf{e}_1, \mathbf{e}_3) + \mathbf{t}$ ($\mathbf{t}$ any translation vector) with numpy, and check whether the origin $\mathbf{0}$ lies in it.

Q21 Use sympy symbolic computation to verify the scalar-multiplication identity axiom $1 \cdot \mathbf{a} = \mathbf{a}$ for a symbolic vector $\mathbf{a}$.

Q23 Generate random 3×3 matrices with numpy, estimate the probability their columns are independent (full rank), and explain why.

Q25 Write a combined script: given any vector group, automatically decide whether it's independent, whether it forms a basis, and the dimension of its span.

Q18 Following Code 1, compute the rank of $\mathbf{B}=[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1+\mathbf{e}_2]$, $\mathbf{C}=[\mathbf{e}_1, \mathbf{e}_2]$, and $\mathbf{D}=[\mathbf{e}_1, -\mathbf{e}_1, \mathbf{e}_1/2]$, and explain each result.

Q20 Use `numpy.linalg.solve()` to check whether vector $\mathbf{b}$ can be written uniquely as a combination of the basis $[\mathbf{e}_1, \mathbf{e}_2]$.

Q22 Write `dim_of_span(vectors)` that uses `np.linalg.matrix_rank()` to compute the dimension of the span of a vector group.

Q24 Visualize the six vector groups in Figure 11 with matplotlib, confirming they all span the same “red-green” plane.

10 Prompts to Explore with AI

1 Using everyday examples, explain the 8 axioms of a vector space to me, especially how a scalar distributes over a vector sum.

2 Prove that if $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ is linearly dependent, removing any one vector leaves the other two independent.

3 Explain why RGB color space isn't strictly a vector space, and identify exactly which axiom fails.

4 Show me, in Python with NumPy, how to check whether a given set of column vectors forms a basis for a vector space.

5 Explain the difference between `span()` and basis: why can a dependent group span a space yet not be called a basis?

10 Prompts to Explore with AI

6 Prove that the column rank of a matrix $\mathbf{A}$ equals its row rank (why $\text{rank}(\mathbf{A})$ is well defined).

7 Explain why “full rank”, “invertible”, “independent columns”, and “nonzero determinant” are all equivalent for a square matrix.

8 Give a concrete example showing that one vector space can have many different bases, and that basis vectors need not be orthogonal.

9 Explain the difference between a vector space and an affine space, and give a real application of affine spaces (e.g. in computer graphics).

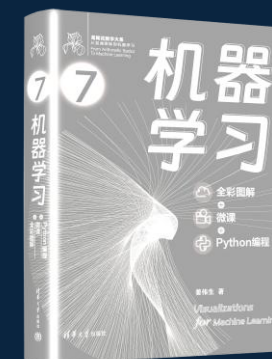
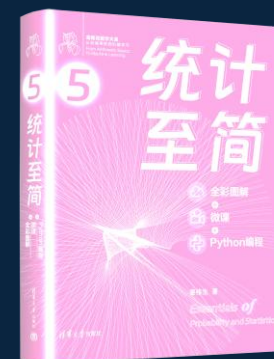
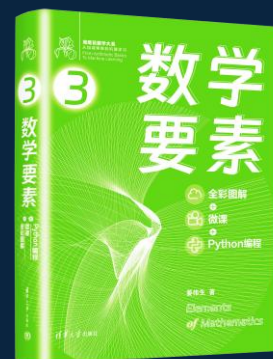
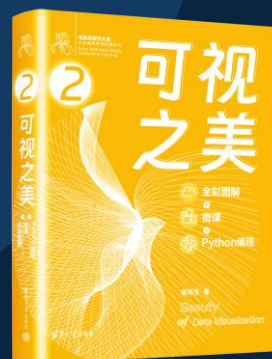
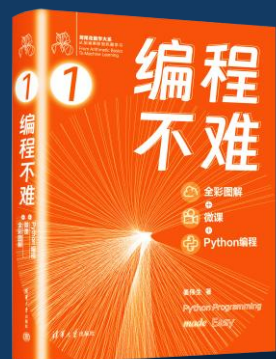
10 Design a Python exercise: given any three 3D vectors, automatically decide whether they are dependent or independent, and visualize the result.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

